searchBAR

Product Overview

SearchBag is a database backed website designed to help SDSU students discover and compare bars in San Diego. Users can search for bars by name, browse available locations, filter based on neighborhood, music type, atmosphere, best days to visit, and ratings, and read reviews written by other users.
The problem that SearchBar addresses is that information about local bars is spread across social media, search engines, review platforms, and word of mouth recommendations. Students may have difficulty finding reliable information about a bar's atmosphere, safety, music, and overall experience. SearchBar is made by students so it keeps fellow students in mind when organizing this information into one centralized website.
SearchBar is meant to provide a simpler way for students to find places to go out in the San Diego area.

Project goals

The main goals of SearchBar are to:

- Allow users to browse a list of San Diego bars.
- Allow users to search for bars by name.
- Allow users to filter bars by relevant characteristics.
- Display detailed information about each bar.
- Display average ratings and written reviews.
- Allow users to submit reviews.
- Demonstrate a functional relational database application.
- Apply database concepts learned in CS514.

Implementation

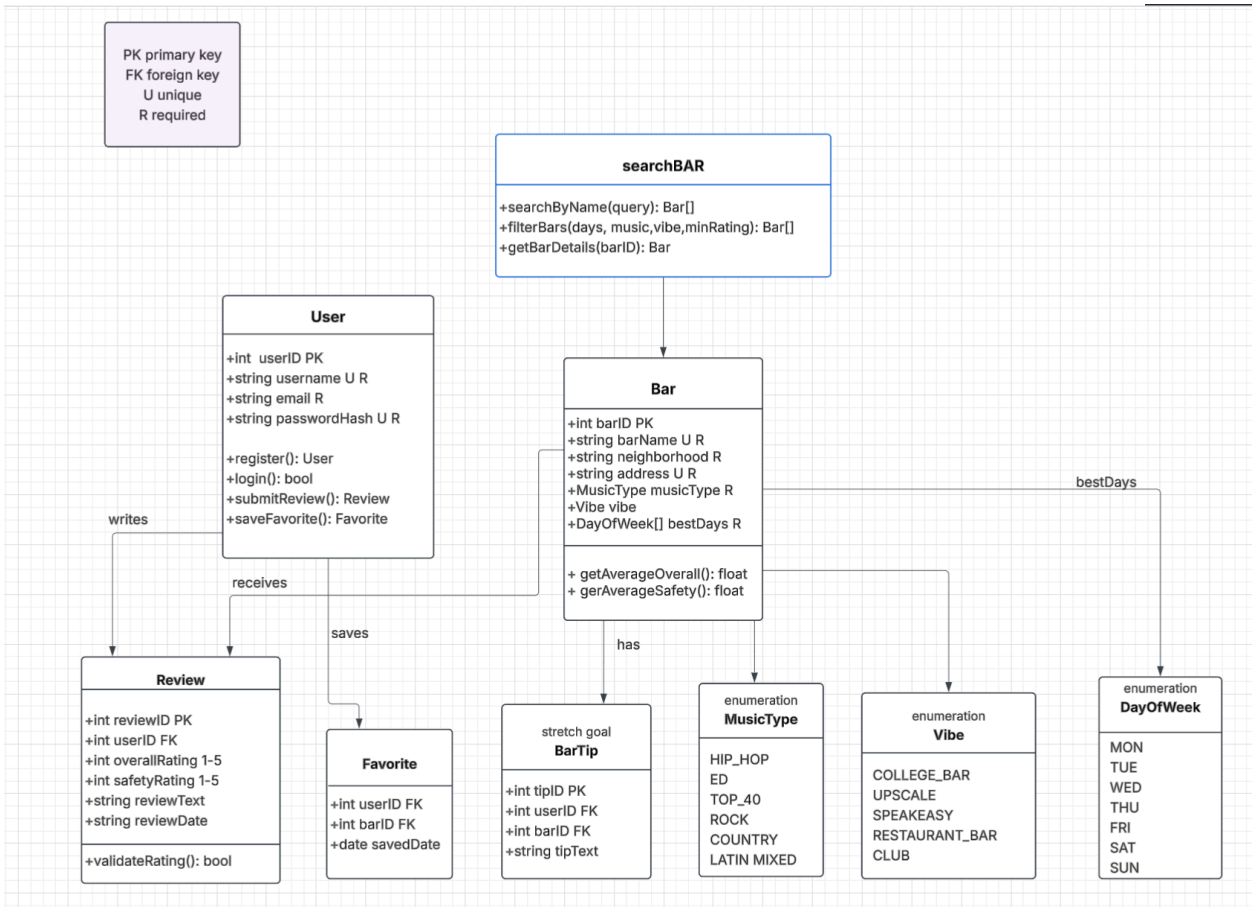
We have two different front end and host implementations of the website currently. They will be presented in class and we will use feedback to pick the more user friendly version. The backend however is through Supabase. It will host our tables and keep our bar information, user information, and reviews. The bar table is currently implemented and the bars are currently displayed on our website. Pictured below.

public.bars

apse sidebar | by bar_id, bar_name, neighborhood... or ask AI | Sort | 1 RLS policy | Role postgres | Insert

bar_id i...	bar_name varchar	neighborhood varchar	address varchar	music_type varchar	vibe
1	Nova Lounge	Gaslamp Quarter	123 Fifth Avenue, San Diego	Live music	Ups
2	The Local	College Area	456 College Avenue, San Diego	DJ nights	Coll
3	Sidewalk Social	Gaslamp Quarter	789 Market Street, San Diego	Mixed	Res

UML Class diagram



Database Operations

SearchBar uses CRUD operations
Create
Users can create:

- A user account
- A reviews
- A favorite bar

Read
Users can view:

- All bars
- Search results
- Filtered results
- Bar details
- Reviews
- Average ratings
- Favorite bars

Update
Users may update

- Their profile information
- Their favorite selections

Authorized user may update:

- Bar information
- Music types
- Neighborhoods
- Atmosphere description

Delete
Users may delete:

- Their own reviews
- Their own favorites

Authorized user may delete:

- Incorrect or duplicate bar listings
- Inappropriate reviews

User Guide

Netlify Implementation - SearchBAR is hosted through Netlify and can be accessed at https://searchbarsd.netlify.app/ without downloading any software. Users can enter a bar name or keyword into the search bar and use the search button to find matching locations. They can also explore bars by filtering them according to location, atmosphere, music type, best days to visit, and ratings. Each bar listing provides details such as its neighborhood, atmosphere, music, best nights, average rating, and user reviews. In later versions, users will be able to create accounts, submit reviews, save favorite bars, and update or remove listings when authorized. The current Netlify version is an early front-end draft, while the Supabase database, login system, reviews, favorites, and complete search functions will be added later. The website is connected to GitHub, so future code pushed to the main branch will automatically update the Netlify website

Cloudflare Implementation

Exploring Bars

The Explore page allows users to search and filter San Diego bars based on their preferences.

Searching for a Bar

1. Select **Explore** from the navigation menu.
2. Enter a bar name, neighborhood, or type of music in the search bar.
3. Click **Search**.
4. The page will display bars that match your search.

For example, searching for **“Gaslamp”** will display bars located in the Gaslamp Quarter.

Using Filters

Users can also select a filter button:

- **All Bars** displays every bar.
- **Gaslamp** displays bars in the Gaslamp Quarter.
- **College Area** displays bars near SDSU.
- **Live Music** displays bars that offer live music.
- **More** filters to come

Viewing Bar Information

Each bar card displays:

- Bar name
- Neighborhood
- Music type
- Atmosphere
- Best day to visit
- Overall rating
- Safety rating

Use this information to compare bars and choose the location that best matches your plans.

Course Concepts Applied

SearchBAR applies several database concepts learned in the course. Entity-relationship modeling is used to identify important entities such as users, bars, neighborhoods, music types, atmospheres, reviews, and favorites, along with the relationships between them. The project uses relational database design by storing these types of information in separate but connected tables. Primary keys uniquely identify records, while foreign keys connect users to their reviews, favorites, and other activities. Normalization helps reduce duplicate information by storing neighborhoods, music types, and atmosphere descriptions separately. SearchBAR also uses CRUD operations because users can create accounts and reviews, read bar information and ratings, update their profiles or favorites, and delete their own reviews or saved locations. SQL joins can combine information from multiple tables, while aggregate functions such as AVG() and COUNT() can calculate ratings and review totals. Constraints such as primary keys, foreign keys, NOT NULL, UNIQUE, and CHECK rules help maintain accurate data. The React and Vite front end will communicate with the Supabase database, while GitHub supports team collaboration and Netlify hosts the deployed application.